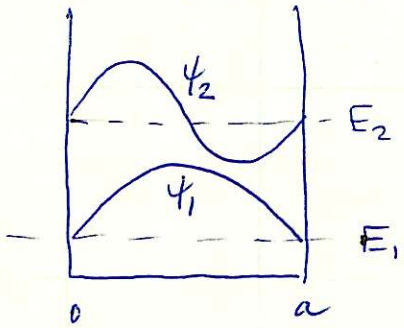
P1



$$\psi_n(x) = \sqrt{\frac{2}{a}}\,\sin\left(\frac{\pi n}{a}x\right)$$

$$P = \int_{\frac{2a}{3}}^{a} dx\,|\psi_2(x)|^2 = \frac{2}{a}\int_{2a/3}^{a} dx\,\sin^2\left(\frac{2\pi x}{a}\right) = \left(\frac{2\pi}{a}x = y\right)$$

$$= \frac{a}{2\pi}\,\frac{2}{a}\int_{4\pi/3}^{2\pi} dy\,\sin^2(y) = \left(\sin^2 y = \frac{1}{2}(1-\cos 2y)\right) =$$

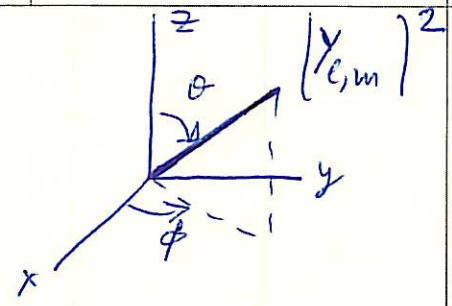
$$= \frac{1}{2\pi}\int_{4\pi/3}^{2\pi} dy\,(1-\cos 2y) = \frac{1}{2\pi}\left[\left(2\pi - \frac{4\pi}{3}\right) - \frac{1}{2}\int_{4\pi/3}^{2\pi} d(\sin 2y)\right] =$$

$$= \frac{1}{2\pi}\left[\frac{2\pi}{3} - \frac{1}{2}\left(\sin 4\pi - \sin\frac{8\pi}{3}\right)\right] = \frac{1}{3} + \frac{1}{4\pi}\sin\frac{8\pi}{3}$$

but $\sin\left(\frac{8\pi}{3}\right) = \sin\left(\frac{9\pi}{3} - \frac{\pi}{3}\right) = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$

$$\Rightarrow \boxed{P = \frac{1}{3} + \frac{\sqrt{3}}{8\pi}}$$

$\boxed{P2}$ $\psi(\vec{r}) = (x+y+z)\, f(r)$



(a) Using Ch. 9 (p. 266)

$$Y_{1,\pm1}(\theta,\phi) = \mp\sqrt{\frac{3}{8\pi}}\, e^{\pm i\phi}\sin\theta$$

$$Y_{1,0}(\theta,\phi) = \sqrt{\frac{3}{4\pi}}\cos\theta \Rightarrow \cos\theta = \frac{z}{r} \Rightarrow Y_{1,0} = \sqrt{\frac{3}{4\pi}}\,\frac{z}{r}$$

$$Y_{1,+1} + Y_{1,-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\,\underbrace{\left(-e^{i\phi} + e^{-i\phi}\right)}_{-2i\sin\phi} = -2i\sqrt{\frac{3}{8\pi}}\,\frac{y}{r}$$

$$Y_{1,+1} - Y_{1,-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\,\underbrace{\left(-e^{i\phi} - e^{-i\phi}\right)}_{-2\cos\phi} = -2\sqrt{\frac{3}{8\pi}}\,\frac{x}{r}$$

where
$$\begin{cases} x = r\cos\phi\sin\theta \\ y = r\sin\phi\sin\theta \\ z = r\cos\theta \end{cases}$$

$$\Rightarrow \frac{x+y+z}{r} = \sqrt{\frac{4\pi}{3}}\,Y_{1,0} - \sqrt{\frac{8\pi}{3}}\,\frac{Y_{1,+1}+Y_{1,-1}}{2i} - \sqrt{\frac{8\pi}{3}}\,\frac{Y_{1,+1}-Y_{1,-1}}{2}$$

$$= \sqrt{\frac{4\pi}{3}}\left[Y_{1,0} + \frac{i}{\sqrt{2}}\left(Y_{1,+1}+Y_{1,-1}\right) - \frac{1}{\sqrt{2}}\left(Y_{1,+1}-Y_{1,-1}\right)\right]$$

$$= \sqrt{\frac{4\pi}{3}}\left(\frac{-1+i}{\sqrt{2}}\,Y_{1,+1} + \frac{1+i}{\sqrt{2}}\,Y_{1,-1} + Y_{1,0}\right)$$

(b) By definition, $\hat{L}_z\, Y_{\ell,m} = \hbar m\, Y_{\ell,m}$

$$\hat{L}^2\, Y_{\ell,m} = \hbar^2 \ell(\ell+1)\, Y_{\ell,m}$$

$\Rightarrow$ we can write $\psi(\vec{r}) = \sqrt{\frac{4\pi}{3}}\, f(r)\cdot r \cdot \left[\frac{-1+i}{\sqrt{2}}\,Y_{1,+1} + Y_{1,0} + \frac{1+i}{\sqrt{2}}\,Y_{1,-1}\right]$

Since $Y_{l,m}$ are normalized $\int_0^{\pi} d\theta \sin\theta \int_0^{2\pi} d\phi \, |Y_{l,m}|^2 = 1$
and independent of each other,

$\psi(\vec{r})$ has to be normalized by dividing by $\frac{1}{\sqrt{4\pi}}$.

$$\psi(\vec{r}) = r f(r) \left(\frac{-1+i}{\sqrt{6}} Y_{1,+1} + \frac{1}{\sqrt{3}} Y_{1,0} + \frac{1+i}{\sqrt{6}} Y_{1,-1} \right)$$

Since $\hat{L}_z Y_{l,m} = \hbar m Y_{l,m}$
we have 3 possibilities:

$L_z = \hbar$ with probability $\left|\frac{-1+i}{\sqrt{6}}\right|^2 = \frac{2}{6} = \frac{1}{3}$

$L_z = 0 \longrightarrow \left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$

$L_z = -\hbar \longrightarrow \left|\frac{1+i}{\sqrt{6}}\right|^2 = \frac{1}{3}$

(c) Only $Y_{1,m}$ appear in $\psi \Rightarrow$

$$\hat{L}^2 \psi = \hbar^2 1(1+1) \psi \Rightarrow \vec{L}^2 = 2\hbar^2$$

corresponding to $l=1$ state.

P3 $H_0 = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$, $H_1 = -\beta x^3$

(a) $E_0^{(0)} = \frac{1}{2}\hbar\omega$

$E_0^{(1)} = \langle \psi_0 | \hat{H}_1 | \psi_0 \rangle = 0$ because $\langle \psi_0 | x^3 | \psi_0 \rangle = 0$

$$E_0^{(2)} = \sum_{k\neq 0}{}' \frac{|\langle \psi_k | \hat{H}_1 | \psi_0 \rangle|^2}{E_0^{(0)} - E_k^{(0)}} = \frac{\beta^2}{\hbar\omega} \sum_{k\neq 0} \frac{|\langle k | x^3 | 0 \rangle|^2}{\frac{1}{2} - (k + \frac{1}{2})}$$

$$= -\frac{\beta^2}{\hbar\omega} \sum_{k\neq 0} \frac{|\langle k | x^3 | 0 \rangle|^2}{k}$$

where $|k\rangle \equiv |\psi_k^{(0)}\rangle$ for harmonic oscillator

Now, $\langle k | x^3 | 0 \rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2} \langle k | (\hat{a} + \hat{a}^+)^3 | 0 \rangle =$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2} \langle k | \left(3|1\rangle + \sqrt{6}|3\rangle\right) = \left(\frac{\hbar}{2m\omega}\right)^{3/2} \left(3\delta_{k,1} + \sqrt{6}\,\delta_{k,3}\right).$$

Because:

$$(a + a^+)^3 = (a + a^+)(aa + a^+a^+ + aa^+ + a^+a) =$$

$$= aaa + aa^+a^+ + aaa^+ + aa^+a + a^+aa + a^+a^+a^+ + a^+aa^+ + a^+a^+a$$

Since $a|0\rangle = 0$, we have

$$(a + a^+)^3 |0\rangle = (aa^+a^+ + aaa^+ + a^+a^+a^+ + a^+aa^+)|0\rangle =$$

Use $\binom{7.33}{7.32}$

$$= aa^+|1\rangle + aa|1\rangle + a^+a^+|1\rangle + a^+a|1\rangle =$$

$$= a\sqrt{2}|2\rangle + a|0\rangle + a^+\sqrt{2}|2\rangle + a^+|0\rangle =$$

$$= \sqrt{2}\cdot\sqrt{2}|1\rangle + 0 + \sqrt{2}\cdot\sqrt{3}|3\rangle + |1\rangle =$$

$$= 3|1\rangle + \sqrt{6}|3\rangle$$

$$\Rightarrow \quad E_0^{(2)} = -\frac{\beta^2}{\hbar\omega} \sum_{k\neq 0} \left(\frac{\hbar}{2m\omega}\right)^3 \frac{\left(3\,\delta_{k,1} + \sqrt{6}\,\delta_{k,3}\right)^2}{k}$$

$$= -\frac{\beta^2}{\hbar\omega} \frac{\hbar^3}{8m^3\omega^3} \left(\frac{9}{1} + \frac{6}{3}\right) = -\frac{\hbar^2\beta^2}{8m^3\omega^4}(11) =$$

$$= -\frac{11}{8} \frac{\hbar^2\beta^2}{m^3\omega^4}$$

(b) The corresponding wave function correction is given by

$$|\psi_0^{(1)}\rangle = \sum_{k\neq 0} |k\rangle \frac{\langle k|\hat{H}_1|0\rangle}{E_0^{(0)} - E_k^{(0)}} =$$

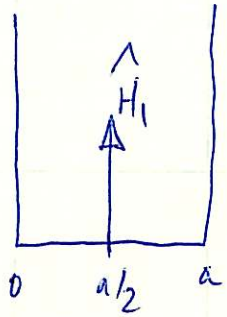
$$= \frac{\beta}{\hbar\omega} \sum_{k\neq 0} |k\rangle \left(\frac{\hbar}{2m\omega}\right)^{3/2} \frac{3\,\delta_{k,1} + \sqrt{6}\,\delta_{k,3}}{k}$$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2} \frac{\beta}{\hbar\omega} \left(3\,|1\rangle + \frac{\sqrt{6}}{3}\,|3\rangle\right).$$

That is, there are contributions from $k=1$ and $k=3$ states.

$\boxed{P4}$ $H_1 = \beta a \varepsilon_1 \delta(x - a/2)$, $\varepsilon_1 \equiv E_1^{(0)}$

(a)



$$E_n^{(0)} = \frac{\hbar^2 \pi^2 n^2}{2ma^2} \quad , \quad \psi_n^{(0)} = \sqrt{\frac{2}{a}} \sin \frac{\pi n x}{a}$$

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle =$$

$$= \frac{2}{a} \int_0^a dx \, (\beta a \varepsilon_1) \sin^2\left(\frac{\pi n x}{a}\right) \delta(x - a/2) =$$

$$= \frac{2}{a} (\beta a \varepsilon_1) \sin^2\left(\frac{\pi n}{2}\right) = \begin{cases} 2\beta\varepsilon_1 & , n = 2k+1 = \underline{\text{odd}} \\ 0 & , n = 2k = \underline{\text{even}} \end{cases}$$

(b) $$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle \psi_k^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

Need $$\langle \psi_k^{(0)} | \hat{H}_1 | \psi_n^{(0)} \rangle = \frac{2}{a} \int_0^a dx \, (\beta a \varepsilon_1) \cdot$$

$$\cdot \sin\left(\frac{\pi k x}{a}\right) \delta(x - a/2) \sin\left(\frac{\pi n x}{a}\right) =$$

$$= 2\beta\varepsilon_1 \sin\left(\frac{\pi k}{2}\right) \sin\left(\frac{\pi n}{2}\right)$$

$$\Rightarrow E_n^{(2)} = (2\beta\varepsilon_1)^2 \sum_{k \neq n} \frac{\sin^2\left(\frac{\pi k}{2}\right) \sin^2\left(\frac{\pi n}{2}\right)}{\frac{\pi^2 \hbar^2}{2ma^2}(n^2 - k^2)} =$$

$$= 4\beta^2 \varepsilon_1 \sum_{k \neq n} \frac{\sin^2\left(\frac{\pi k}{2}\right) \sin^2\left(\frac{\pi n}{2}\right)}{n^2 - k^2}$$

where I used that $\frac{\pi^2 \hbar^2}{2ma^2} = E_1^{(0)} = \varepsilon_1$.

(a) Observe that $E_n^{(2)} = 0$ for <u>n = even</u>

because then $\sin\left(\frac{\pi}{2} n\right) = 0$.

This will hold to all orders of the perturbation theory.

(b) For $n=1$, $\sin^2\left(\frac{\pi n}{2}\right) = 1$, and we need

$$J = \sum_{k \neq 1} \frac{\sin^2\left(\frac{\pi k}{2}\right)}{1-k^2} = (k = 2p+1) =$$

$$= \sum_{p=1}^{\infty} \frac{1}{1-(2p+1)^2} = -\sum_{p=1}^{\infty} \frac{1}{2p\cdot(2p+2)} =$$

$$= -\sum_{p=1}^{\infty} \left(\frac{1}{2p} - \frac{1}{2p+2}\right)\cdot\frac{1}{2} = \left(\text{observe } \sum_{p=1}^{\infty} \frac{1}{p+1} = \sum_{p=2}^{\infty} \frac{1}{p}\right)$$

$$= -\frac{1}{4} \sum_{p=1}^{\infty} \left(\frac{1}{p} - \frac{1}{p+1}\right) = \frac{1}{4}\left(\sum_{p=2}^{\infty} \frac{1}{p}\right) - \frac{1}{4}\left(\sum_{p=1}^{\infty} \frac{1}{p}\right)$$

$$= \frac{1}{4}\left\{\underbrace{\left(\sum_{p=1}^{\infty} \frac{1}{p} - 1\right)}_{\sum_{p=2}^{\infty} \frac{1}{p}} - \sum_{p=1}^{\infty} \frac{1}{p}\right\} = -\frac{1}{4}$$

Thus $E_{n=1}^{(2)} = 4\beta^2 \varepsilon_1 \left(-\frac{1}{4}\right) = -\beta^2 \varepsilon_1$

$$\Rightarrow E_{n=1} = E_{n=1}^{(0)} + E_{n=1}^{(1)} + E_{n=1}^{(2)} =$$

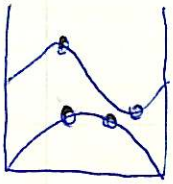
$$= E_{n=1}^{(0)} \left(1 + 2\beta - \beta^2\right)$$

$\boxed{P5}$ B. NUS $\boxed{G\ 6.3}$ $V(x_1, x_2) = -a V_0\, \delta(x_1 - x_2)$

Two identical bosons in infinite square well

(a) $\psi_{gs}(x_1 x_2) = \psi_1(x_1)\psi_1(x_2)$ $\qquad E_{gs} = E_0 = 2 \cdot \frac{\pi^2\hbar^2}{2ma^2}$

$$\psi_{ex}(x_1 x_2) = \frac{1}{\sqrt{2}}\left(\psi_1(x_1)\psi_2(x_2) + \psi_1(x_2)\psi_2(x_1)\right)$$



$$\Rightarrow E_1 = \frac{\pi^2\hbar^2}{2ma^2}(1+2^2)$$

(b) $E_0^{(1)} = \langle \psi_1(x_1)\psi_1(x_2) | \hat{V} | \psi_1(x_1)\psi_1(x_2)\rangle$

$$= -aV_0 \int_0^a dx_1 dx_2\ \delta(x_1 - x_2)\, |\psi_1(x_1)|^2\, |\psi_1(x_2)|^2$$

$$= -aV_0 \int_0^a dx\ |\psi_1(x)|^4$$

$$= -aV_0 \left(\frac{2}{a}\right)^{\frac{1}{2}\cdot 4} \int_0^a dx\ \sin^4\left(\frac{\pi x}{a}\right) = -aV_0\left(\frac{2}{a}\right)^2 \frac{a}{\pi}\int_0^\pi dy\ \sin^4(y)$$

$$\frac{\pi x}{a} = y$$

$$= -\frac{4V_0}{\pi}\frac{3\pi}{8} = -\frac{3}{2}V_0.$$

$$I_1 = \int_0^\pi dx\ \sin^4 x = \frac{1}{4}\int_0^\pi dx\ (1-\cos 2x)^2 = \frac{1}{4}\int_0^\pi (1 - 2\cos 2x + \cos^2 2x)\,dx$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$$

$$= \frac{1}{4}\left(\pi - \int_0^{2\pi} dt\ \cos t + \frac{1}{2}\underbrace{\int_0^{2\pi} dt\ \cos^2 t}_{\pi}\right) = \frac{1}{4}\left(\pi + \frac{\pi}{2}\right) = \frac{3\pi}{8}$$

$$E_1^{(1)} = \frac{1}{2}\int dx_1 dx_2\ \left(\psi_1(x_1)\psi_2(x_2) + \psi_1(x_2)\psi_2(x_1)\right)\left(-aV_0\,\delta(x_1 - x_2)\right)\cdot$$

$$\cdot\left(\psi_1(x_1)\psi_2(x_2) + \psi_1(x_2)\psi_2(x_1)\right)$$

$$= -\frac{aV_0}{2}\int dx\ \left(\psi_1(x)\psi_2(x) + \psi_1(x)\psi_2(x)\right)^2 = -2aV_0\int_0^a dx\ |\psi_1(x)|^2|\psi_2(x)|^2$$

$$E_1^{(1)} = -\frac{2aV_0}{\pi}\left(\frac{2}{a}\right)^2 \int_0^a dx\, \sin^2\left(\frac{\pi x}{a}\right) \sin^2\left(\frac{2\pi x}{a}\right) =$$

$$= -\frac{2aV_0}{\pi}\,\frac{4}{a^2}\,\frac{a}{\pi} \int_0^\pi dy\, \sin^2 y\, \sin^2 2y =$$

$$= -\frac{8}{\pi} V_0 \int_0^\pi dy\, \sin^2 y\; 4\sin^2 y \cos^2 y = -\frac{32}{\pi} V_0 \int_0^\pi dy\, \sin^4 y\,(1-\sin^2 y)$$

$$= -\frac{32}{\pi} V_0 \left[\frac{3\pi}{8} - \int_0^\pi dy\, \sin^6 y\right]$$

$$I_2 = \int_0^\pi dy\, \sin^6 y = \int_0^\pi dy \left(\frac{(1-\cos 2y)}{2}\right)^3 = \frac{1}{8}\int_0^\pi dy\,(1-2\cos 2y+\cos^2 2y)\cdot$$

$$\cdot(1-\cos 2y) = \frac{1}{8}\int_0^\pi dy\,(1-2\cos 2y+\underline{\cos^2 2y}-\cos 2y+\underline{2\cos^2 2y}-\cos^3 2y)$$

$$= \frac{1}{8}\int_0^\pi dy\,(1-3\cos 2y+3\cos^2 2y-\cos^3 2y) = \cancel{y=2t}\ (t=2y) =$$

$$= \frac{1}{16}\int_0^{2\pi} dt\,(1-3\cos t+3\cos^2 t-\cos^3 t) =$$

$$= \frac{1}{16}(2\pi+3\pi) = \frac{5\pi}{16}$$

$$\int_0^{2\pi} dt\, \cos^3 t = \int d(\sin t)\,(1-\sin^2 t) = 0$$

$$\Rightarrow E_1^{(1)} = -\frac{32}{\pi} V_0 \left(\frac{3\pi}{8} - \frac{5\pi}{16}\right) = -\frac{32V_0}{\pi}\,\frac{\pi}{16} = -2V_0$$

The result

$$\begin{cases} E_0 = \dfrac{\pi^2\hbar^2}{ma^2} - \dfrac{3V_0}{2} & \leftarrow \text{ground state} \\[2ex] E_1 = \dfrac{5\pi^2\hbar^2}{2ma^2} - 2V_0 & \leftarrow 1^{ST}\text{ excited state} \end{cases}$$